

Trader Performance vs Market Sentiment Analysis

A Data-Driven Analysis using Bitcoin Fear & Greed Index and Trader Data

1. Introduction

The report examines the connection between the Bitcoin market sentiment (Fear, Greed, and extreme variants) and the performance of the traders based on historical trading data. The goal is to find trends in trader behavior and give practical information to be more intelligent in trading.

2. Dataset Description

Two datasets were taken:

- Market Sentiment (Fear and Greed Index) Dataset.

Hypoliquid Data: This is the data that is traded historically and is known as historical data.

These datasets were combined with date to understand the effect of sentiment on trading decisions and performance.

3. Analysis & Visual Insights



Figure: Profit Distribution by Market Sentiment

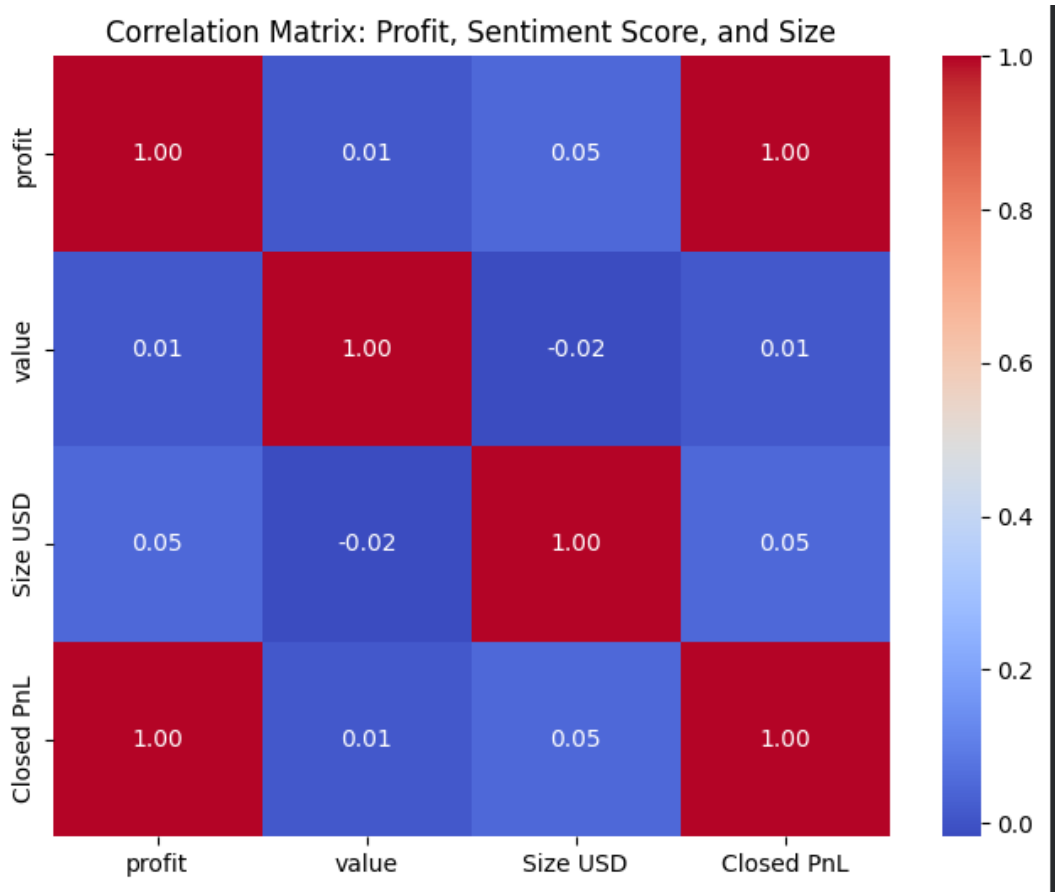


Figure: Correlation Matrix

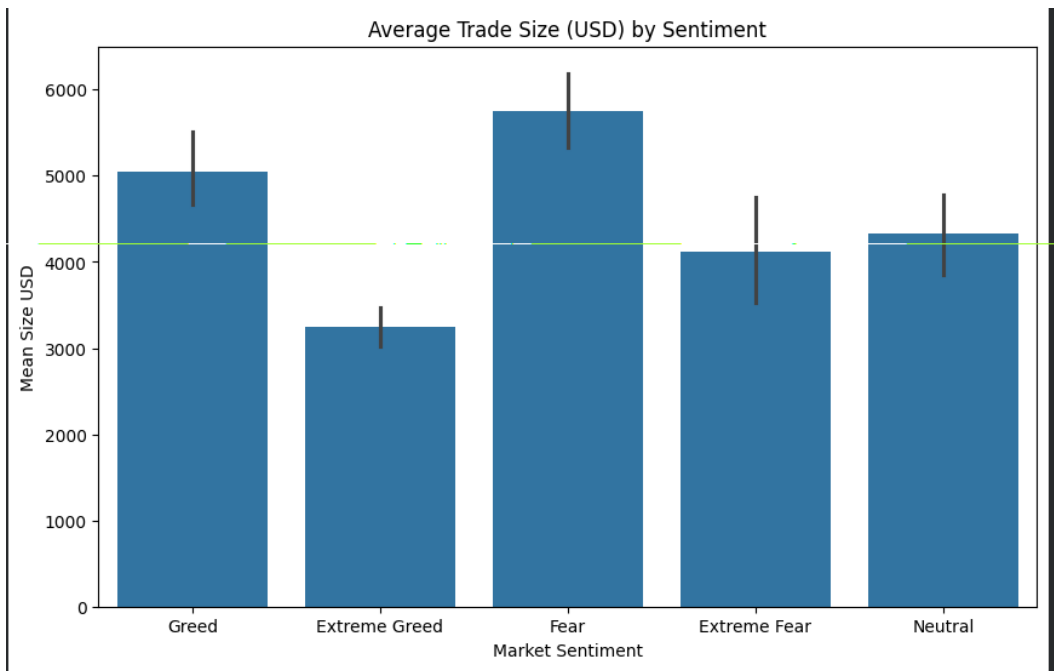


Figure: Average Trade Size by Sentiment

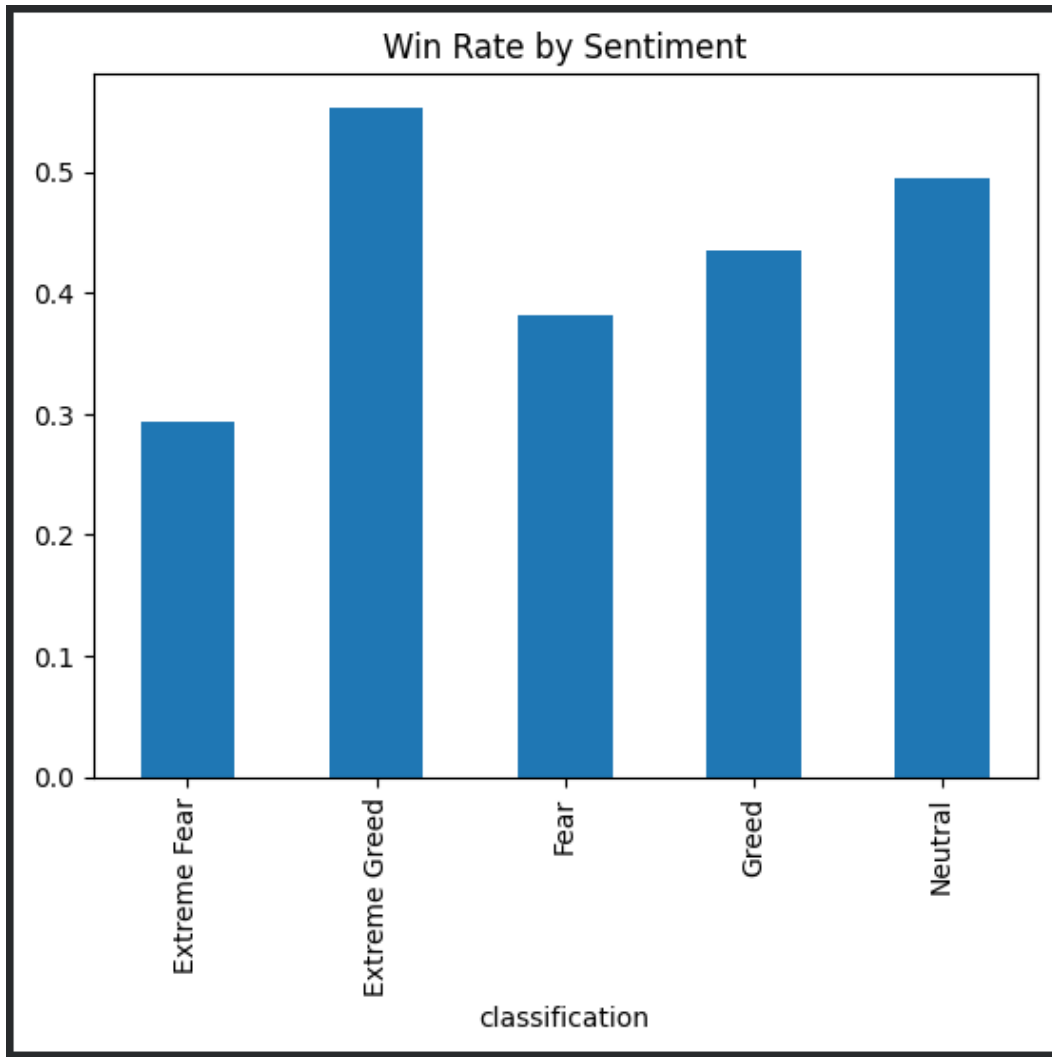


Figure: Win Rate by Sentiment

4. Key Insights

- There is a very high volatility in trader profitability at all sentiment phases.
- The win rate is the highest with Extreme Greed (~55%), and the lowest with Extreme Fear (~29%).
- Traders: When there is Fear, traders assume bigger positions, which are a sign of risk-seeking behaviour.
- Sentiment and profit have low correlation indicating that performance is affected by outside forces.

5. Behavioral Insights

The traders have emotional patterns of decision making:

- Fear > Bigger trades but less successful.

Greed: Increased win rate but volatile profits.

This brings out the influence of psychology more than the naked market indicators.

6. Conclusion

The sentiment in the market has a great influence on the traders but the profitability is not ensured. The trading requires disciplined trading and management of risks.

7. Tools & Technologies

Data analysis and visualization were done in Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn).